



CAPSTONE PROJECT PROPOSAL

Project Title	Data-Driven Analysis of Global AI Compute Dependency and Arab AI Opportunities
Track	Junior Data Analysis
Job Profile	Data Analyst / BI Developer
Program Name	Nano Degree
Team Name	The AI Compass
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Supervision Committee	supervisor: Prof. Dr. Rania Elgohary Track Head: Dr. Ahmed Zakaria Academic Team: Dr. Yasser Ali instructor: Dr: Bassam Elshoura , YAT Company
Submission Date	September 2026

Declaration and Certificate of Originality

We, the undersigned team members, declare that this capstone project proposal is our original work and has been prepared for academic evaluation purposes. Any external datasets, tools, libraries, frameworks, or resources used in the project will be clearly acknowledged within this final project documentation according to the committee requirements.

We confirm that the proposed project idea, implementation plan, and expected deliverables are developed by the team and that the project will follow ethical, legal, and professional standards related to data, security, software development, and technology usage.

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Dedication

We proudly dedicate this capstone project to the Digilians (Digital Pioneers) Initiative, the Egyptian Military Academy, and the Ministry of Communications and Information Technology (MCIT). Your visionary leadership and unwavering commitment to empowering the next generation of Egyptian tech leaders have provided us with an exceptional platform to learn, innovate, and excel in the world of data analytics.

We also dedicate this work to our esteemed mentors, instructors, and the supervision committee. Your technical guidance, constructive feedback, and endless patience have been the cornerstone of our academic and professional growth throughout this intensive journey.

To our families and loved ones—thank you for your boundless encouragement, sacrifices, and belief in our potential. This milestone is as much yours as it is ours. We also dedicate this success to one another as a team; our shared late nights, collaborative spirit, and dedication transformed a challenging idea into a tangible reality.

Finally, this project is dedicated to the vision of a digitally transformed and technologically sovereign Egypt. May our efforts in analyzing the global AI ecosystem contribute to Egypt Vision 2030, inspiring future generations to build data-driven solutions that secure our nation's place as a digital leader in the Arab world and beyond.

Acknowledgement

"The team would like to express our deepest gratitude and appreciation to the program management of the Digilians (Digital Pioneers) Initiative, the Egyptian Military Academy, and the Ministry of Communications and Information Technology (MCIT) for providing this exceptional learning platform and for their continuous support in fostering digital transformation.

We extend our sincere thanks to our Supervision Committee, particularly

supervisor: Prof. Dr. Rania ELgohary

Track Head: Dr. Ahmed Zakaria

Academic Team: Dr. Yasser Ali

instructor: Dr. Bassam Elshoura & YAT Company.

, for their invaluable technical guidance, constructive feedback, and steadfast mentorship throughout the project development process. Their expertise in data analytics and business intelligence was instrumental in shaping the success of this capstone project.

Finally, we would like to acknowledge the academic team members and external contributors whose resources and insights significantly enriched the analytical depth of our AI compute ecosystem research."

Abstract

This project proposes the development of a comprehensive Business Intelligence (BI) solution to analyze the global Artificial Intelligence (AI) ecosystem, focusing on compute dependency, market concentration, and Arab AI opportunities. By tracing the historical evolution of AI systems—from their early inception to the highly advanced models of 2026—the project maps exponential compute trends and technological shifts. The proposed system aims to support policymakers, researchers, and tech investors by providing data-driven insights into AI infrastructure and model-development capabilities. Furthermore, it identifies the most widespread and rapidly adopted AI domains, guiding stakeholders on optimal areas for strategic tech investments.

The solution utilizes Microsoft Power BI, Power Query (M Language), and DAX to process and visualize extensive datasets. The expected output is an interactive dashboard featuring analytical pages, calculated measures, and dimensional hierarchies designed to track global AI scale, openness, and regional readiness.

Crucially, this project is deeply aligned with Egypt Vision 2030 and its strategic pillar of building a competitive, balanced, and knowledge-based economy. As Egypt accelerates its national digital transformation through capacity-building programs and sovereign cloud expansions in 2026, there is an urgent national mandate to localize advanced technologies. By providing an evidence-based roadmap, this solution empowers Egyptian decision-makers to transition the local ecosystem from mere infrastructure consumers to active AI model developers. Ultimately, it highlights how optimizing digital investments and fostering local talent can secure Egypt's technological sovereignty and establish it as a leading, globally competitive hub for AI innovation in the region.

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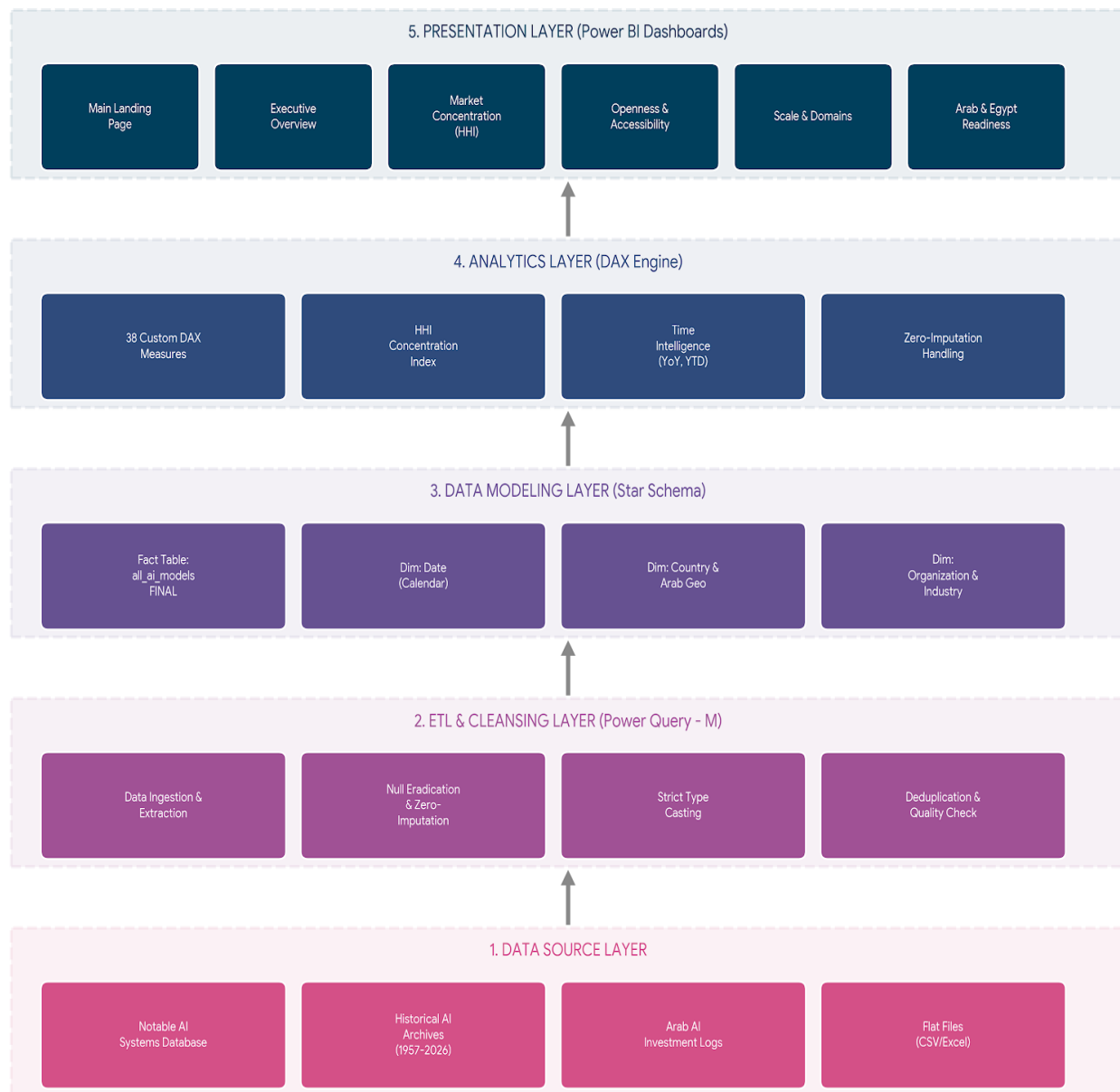
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Figure 1: End-to-End Project Methodology & System Architecture



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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
BI	Business Intelligence
DAX	Data Analysis Expressions
HHI	Herfindahl-Hirschman Index
ETL	Extract, Transform, Load
LLM	Large Language Model
MCIT	Ministry of Communications and Information Technology
KPI	Key Performance Indicator
YoY	Year-over-Year
YTD	Year-to-Date
CAGR	Compound Annual Growth Rate
UI / UX	User Interface / User Experience

List of Symbols

Symbol	Meaning / Description
Σ	Summation
N	Total number of AI models
%	Percentage
HHI	Herfindahl-Hirschman Index
CAGR	Compound Annual Growth Rate
s_i	Market share of a specific country/organization

1. Executive Summary

- Executive Summary Artificial Intelligence (AI) has rapidly transcended its origins as a niche field of computer science to become the foundational infrastructure of the global digital economy. Today, advanced AI systems demand unprecedented computational power, specialized semiconductor chips, and massive capital investments. This global shift has created a new paradigm where technological sovereignty is directly tied to AI compute capacity. This capstone project proposes a data-driven Business Intelligence (BI) platform designed to meticulously analyze global AI compute dependency and highlight emerging AI opportunities, particularly within the Arab region and Egypt.

The scope of this project extends beyond basic data visualization; it represents a comprehensive engineering effort to build a robust semantic data model capable of answering complex geopolitical and economic questions regarding AI ownership and open-source accessibility. By consolidating historically fragmented records into a unified, highly optimized data fabric, this project shifts organizational capabilities from reactive auditing to proactive, evidence-based governance.

Developed as part of the Digilians (Digital Pioneers) Initiative, this solution aligns perfectly with Egypt's national mandate for digital transformation. Through the deployment of 38 custom Data Analysis Expressions (DAX) and a highly normalized Star Schema architecture within Microsoft Power BI, this analytical engine provides unprecedented visibility into the Herfindahl-Hirschman Index (HHI) for AI market concentration, open-source adoption rates, and regional AI readiness. Ultimately, this executive dashboard empowers decision-makers to navigate the complexities of the AI ecosystem, optimizing national investments and fostering local innovation

2. Problem Statement

Currently, national decision-makers, technology investors, and strategic planning institutions within the Arab region—and specifically Egypt—face critical challenges related to the extreme centralization of the global Artificial Intelligence (AI) ecosystem. A highly restricted number of multinational corporations and sovereign states currently monopolize advanced AI compute infrastructure, foundation models, and the vast proprietary datasets required for machine learning.

This severe concentration of technological power causes long-term strategic vulnerabilities, directly affecting local stakeholders who struggle to foster indigenous digital innovation and secure technological sovereignty. The negative effects cascade into economic dependency, where developing ecosystems risk remaining perpetual consumers of imported AI infrastructure rather than evolving into active, competitive creators.

On an operational and analytical level, stakeholders attempting to monitor and respond to this landscape are crippled by severe data silos and fragmentation. Historical records tracking AI model releases, parameter scales, and compute costs (spanning from 1957 to 2026) are scattered across disparate CSV files, academic repositories, and unstructured logs.

Existing analytical approaches are severely limited because they rely on manual report generation and disjointed spreadsheet calculations. This manual methodology introduces several critical limitations:

- **Slow Decision Velocity:** Compiling global AI metrics manually takes weeks, rendering the insights outdated by the time they reach executive leadership.
- **Human Error and Analytical Inconsistency:** Manually calculating complex economic metrics, such as the Herfindahl-Hirschman Index (HHI) for market concentration or Compound Annual Growth Rates (CAGR), introduces unacceptable margins of error that can misguide national investment strategies.
- **Zero Predictive Visibility:** Organizations are forced to react to historical audits rather than proactively anticipating forward-looking trends, such as the exponential scaling of model parameters or the sudden emergence of Arabic-focused Large Language Models (LLMs).

Therefore, this capstone project proposes an enterprise-grade Business Intelligence (BI) technical solution to definitively resolve these bottlenecks. By engineering a highly optimized Star Schema and deploying a suite of 38 advanced Data Analysis Expressions (DAX), this project will consolidate fragmented data into a single, authoritative semantic layer. This solution will dramatically improve strategic visibility, support policymakers with sub-second, interactive reporting, and provide the expected benefit of proactive,

evidence-based governance—ensuring national investments are accurately aligned with the digital transformation mandates of Egypt Vision 2030.

3. Project Objectives

To effectively address the centralization of the global AI ecosystem and deliver actionable insights, this capstone project is driven by the following clear and measurable objectives:

1. **Analyze the selected problem and define system requirements:** Analyze the global AI compute dependency landscape to define rigorous business and technical requirements. This includes establishing strict data governance rules for centralizing and processing over 3,500 historical and modern AI model records while resolving extreme data fragmentation.
2. **Design a suitable technical solution:** Design a robust, enterprise-grade data architecture. This involves engineering a highly optimized Star Schema model and establishing resilient ETL (Extract, Transform, Load) pipelines utilizing Power Query (M Language) to execute advanced data cleansing and Zero-Imputation strategies.
3. **Develop a prototype, application, model, dashboard, platform, pipeline, or security mechanism:** Develop a comprehensive, interactive Business Intelligence (BI) dashboard using Microsoft Power BI. This analytical platform will be powered by a sophisticated engine comprising 38 custom Data Analysis Expressions (DAX) distributed across 6 dedicated, executive-level report pages.
4. **Test the proposed solution using suitable evaluation methods:** Test and validate the proposed BI solution through rigorous data profiling and quality checks. This includes verifying the mathematical accuracy of complex DAX calculations (such as the Herfindahl-Hirschman Index and Compound Annual Growth Rates) and optimizing the VertiPaq engine to guarantee sub-second visual refresh rates.
5. **Provide meaningful outputs such as insights, reports, alerts, predictions, automation, or user services:** Provide real-time, meaningful executive insights regarding global AI market concentration, open-source adoption trends, and technological scaling. Specifically, the project aims to generate automated reports evaluating Arab AI readiness and Egypt's position to support data-driven policy-making and strategic investments.
6. **Demonstrate the practical value of the solution through a final demo:** Demonstrate the practical and strategic value of the analytical platform through a comprehensive final presentation to the Digilians initiative supervision committee, showcasing how this BI tool directly empowers national digital transformation and aligns with Egypt Vision 2030.

4. Project Scope

The scope of this capstone project is strictly focused on the end-to-end data lifecycle, from raw data extraction to the deployment of an executive Business Intelligence dashboard. It establishes clear boundaries to ensure the project remains focused on delivering actionable analytics regarding the global AI ecosystem. The project will include:

- **Problem analysis and requirements definition:** Conducting a comprehensive analysis of the global AI compute dependency and data fragmentation challenges. This includes defining strict functional requirements for the BI solution (e.g., automated HHI calculation) and non-functional requirements (e.g., sub-second visual refresh rates and data accuracy).
- **Dataset collection or system input definition:** Ingesting the central "Notable AI Systems Database," which comprises over 3,500 historical and modern AI model records spanning from 1957 to 2026. The scope covers the extraction of key quantitative inputs (parameters, citations) and categorical variables (organizations, domains, accessibility).
- **System design and architecture:** Designing a robust data architecture utilizing a highly normalized Star Schema. This encompasses the conceptualization of one central Fact Table and multiple supporting Dimension Tables (Date, Country, Organization, Domain) mapped with single-direction relationships to optimize the VertiPaq engine.
- **Implementation of the core features:** Executing rigorous ETL (Extract, Transform, Load) pipelines via Power Query (M Language) for data cleansing, deduplication, and Zero-Imputation logic. Furthermore, the core implementation includes engineering 38 custom Data Analysis Expressions (DAX) categorized into 8 logical folders to compute advanced time intelligence and market concentration metrics.
- **Model, pipeline, platform, application, device integration, or security solution development:** Developing and deploying a comprehensive 6-page interactive Business Intelligence platform using Microsoft Power BI. The scope includes designing the User Interface/User Experience (UI/UX) for the Main Landing Page, Executive Overview, Concentration, Openness, Scale, and Arab & Egypt Readiness pages.
- **Testing, evaluation, documentation, and final demo:** Testing the mathematical validity of the DAX calculations against raw source files, profiling data quality to ensure the successful handling of null values, and producing extensive technical

documentation. The scope culminates in delivering a final, functional demonstration to the Digilians initiative evaluation committee.

5. Target Users and Stakeholders

Stakeholder	Role / Benefit
End Users	National decision-makers, tech investors, and AI researchers who will use the Power BI dashboard directly to monitor global compute trends, analyze HHI market concentration, and identify open-source AI investment opportunities
Organization / Institution	MCIT, Digilians Initiative, and strategic planning institutions benefiting from evidence-based situational awareness to align national AI capabilities with Egypt Vision 2030 and achieve technological sovereignty.
Admin / Supervisor	BI Administrators and Data Governance Officers who monitor the dashboard's health, manage Power BI workspace access, and oversee the scheduled refresh of the underlying dataset.
Technical Team	Data Analysts, BI Developers, and Data Engineers responsible for maintaining the Star Schema data model, updating the Power Query (M Language) ETL pipelines, and optimizing the 38 DAX measures for future requirements.
Society / Community	The Egyptian tech ecosystem and local AI talent who benefit indirectly from the project's impact in driving national digital transformation, fostering local digital innovation, and shifting the ecosystem towards a knowledge-based economy.

6. Proposed Solution

The proposed solution is an enterprise-grade Business Intelligence (BI) platform and automated analytical pipeline that aims to solve the severe data fragmentation and lack of strategic visibility regarding global AI compute dependency. By using Microsoft Power BI, Power Query (M Language), and advanced Data Analysis Expressions (DAX), the solution establishes a centralized and authoritative semantic layer. The system will receive input from the comprehensive "Notable AI Systems Database" (comprising over 3,577 historical and modern records), process it through rigorous ETL data cleansing, deduplication, and a strict Zero-Imputation strategy, and generate outputs such as a 6-

page interactive executive dashboard, automated Herfindahl-Hirschman Index (HHI) market concentration reports, and regional Arab AI readiness metrics.

At its core, the architecture relies on a highly optimized Star Schema—consisting of a central Fact Table (`all_ai_models_FINAL`) surrounded by dimensional tables for Date, Country, Organization, and Domain—to ensure sub-second query performance via the VertiPaq engine. By replacing manual, spreadsheet-based audits with this automated analytical engine powered by 38 custom DAX measures, the solution directly addresses the operational bottlenecks of slow decision velocity and calculation errors. Ultimately, this comprehensive output empowers national decision-makers and tech investors to proactively identify open-source opportunities and transition the local ecosystem from mere infrastructure consumers to active, competitive AI developers.

7. System Features

Feature	Description
User Interface	Provides an intuitive, interactive environment within Power BI, featuring synchronized slicers, dynamic tooltips, and seamless cross-filtering across all 6 analytical pages.
Data Input / Upload	Utilizes Power Query (M Language) to extract and ingest over 3,577 AI model records, applying strict ETL protocols including deduplication and Zero-Imputation.
Processing Module	Relies on a highly optimized Star Schema data model and the VertiPaq engine to manage 1-to-many relationships and compress data for sub-second query performance.

AI / Analytics Module	Powered by a robust engine of 38 custom Data Analysis Expressions (DAX) to compute advanced metrics such as the Herfindahl-Hirschman Index (HHI) and Time Intelligence (CAGR, YoY).
Security Module	Implements Power BI Workspace access controls and role-based sharing protocols to ensure that strategic insights and executive dashboards are securely distributed.
Dashboard	Displays a comprehensive 6-page visual narrative (Overview, Concentration, Openness, Scale, Arab Readiness) to present KPIs, geographic maps, and trends to stakeholders.
Reporting Module	Automates the generation of strategic executive summaries and allows users to export underlying data and visuals for external policy-making documentation.
Notification / Alert System	Leverages Power BI Service capabilities to configure data-driven alerts, notifying administrators when specific KPI thresholds (e.g., market concentration shifts) are breached.

8. Technical Approach

Given that this project falls under the **Data Analytics and Business Intelligence** track, the technical approach follows a rigorous, end-to-end Data Engineering and Analytics pipeline. The methodology is structured into five sequential phases:

Phase 1: Data Extraction & Ingestion The project initiates by connecting to the "Notable AI Systems Database," extracting over 3,500 historical and modern AI model records. Microsoft Power BI's native connectors are utilized to ingest these flat files and unstructured logs into the staging environment.

Phase 2: ETL & Data Cleansing (Power Query - M Language) To ensure data integrity before modeling, extensive Extract, Transform, and Load (ETL) operations are executed:

- **Deduplication & Type Casting:** Removing duplicate rows to prevent artificially inflated market density metrics, and strictly defining data types (e.g., Dates, Int64, Decimals) for engine stability.

- **Zero-Imputation Strategy:** A critical engineering step to handle empty cells (Nulls) in quantitative columns like 'Parameters' and 'Citations'. To prevent DAX iterator functions from failing or dropping rows, all nulls are imputed with '0'. Secondary Boolean flags are created to distinguish true zeros from imputed ones, preserving the accuracy of statistical medians.

Phase 3: Semantic Data Modeling (Star Schema) The cleansed flat table is decomposed into a highly normalized **Star Schema** to maximize the VertiPaq engine's column-store compression.

- **Fact Table:** A central `all_ai_models_FINAL` table is established to hold all quantitative measures and foreign keys.
- **Dimension Tables:** Dedicated tables for Date (Calendar), Country, Organization, and Domain are built.
- **Relationships:** Single-direction, 1-to-many relationships are configured to prevent circular dependency overhead and ensure accurate cross-filtering.

Phase 4: Analytics Engine (DAX Development) Instead of relying on basic aggregations, the analytical core is powered by 38 custom Data Analysis Expressions (DAX) categorized into logical folders. This includes writing complex logic for Time Intelligence (e.g., YoY growth, Moving Averages) and programming the Herfindahl-Hirschman Index (HHI) to dynamically calculate market concentration across different years and geographies.

Phase 5: Visualization & Presentation (UI/UX) The final phase translates the semantic model into a 6-page interactive executive dashboard. The UI/UX is designed with customized themes, synchronized slicers, drill-through capabilities, and dynamic tooltips, ensuring that complex data narratives (such as Arab AI readiness and global openness) are easily digestible for non-technical policymakers.

8.1 Input / Data Sources

For this analytical project, the data inputs are carefully selected to ensure comprehensive coverage of the historical and current AI landscape. Based on the project scope, the inputs include:

- **Public Datasets:** The primary data source is the "Notable AI Systems & Machine Learning Trends Database" (originally curated by Epoch AI). This is a globally recognized public dataset tracking AI model developments from 1957 to 2026.
- **Tabular Data:** The data is ingested as structured tabular records (CSVs/Excel) containing both

quantitati

ve metrics (e.g., parameter counts, training compute, citations) and categorical information (e.g., developer organization, domain, country of origin).

- **Collected Data (Categorical Mapping):** Supplementary geographic and domain mapping data was collected and integrated to accurately flag "Arab Models" and evaluate Egypt's specific regional readiness within the broader global dataset.

8.2 Data / Input Preparation

- **. Initial State of the Raw Data:** At the beginning of the project, the primary dataset (Epoch AI) was ingested as a highly fragmented and unstructured flat file containing 3,577 records of AI models spanning from 1957 to 2026. The raw data presented several critical challenges:
DOCX +
- **Missing Values (Nulls):** Significant gaps existed in crucial quantitative columns. For instance, only 2,328 out of 3,577 models had recorded parameter sizes, and 835 models had no recorded accessibility data.
PDF +
- **Inconsistent Formatting:** Date columns lacked standardized formats, and categorical fields (such as Developer Organizations and Domains) contained un-normalized text variations that hindered accurate grouping.
- **Flat Structure:** The data was entirely denormalized (a single massive table), which is highly inefficient for BI analytical engines.
- **The End-to-End Journey:** To transform this raw data into an executive dashboard, the team executed a rigorous data engineering and analytics pipeline:
- **Step 1: Extraction & Cleansing (Power Query):** Ingested the raw CSV/Excel files and applied aggressive cleansing. We implemented a "Zero-Imputation" strategy to replace nulls with zeros (using Boolean flags to preserve statistical accuracy) and standardized all text and date fields.
DOCX
- **Step 2: Data Modeling (Star Schema):** Decomposed the massive flat file into a high-performance Star Schema, establishing a central Fact Table (all_ai_models FINAL) connected via 1-to-many relationships to dedicated Dimension tables (Date, Country, Organization, Industry).
DOCX +

- Step 3: Analytics (DAX Engine): Engineered 38 custom DAX measures categorized into distinct folders to dynamically calculate complex metrics such as the HHI market concentration and Year-over-Year growth.
DOCX +
- Step 4: UI/UX & Visualization: Translated the underlying data model into a 6-page interactive Power BI dashboard and integrated the findings into a dedicated local website for seamless stakeholder presentation.
DOCX +.

Rank	Model	Benchmark	Score	Unit	Source	Source URL
1	Applied-Innovation-Center/Karnak	Average	79.29 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
2	Applied-Innovation-Center/Karnak	ALiGhafa	79.01 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
3	Applied-Innovation-Center/Karnak	ArabicMMLU	79.13 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
4	Applied-Innovation-Center/Karnak	EXAMS	63.31 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
5	Applied-Innovation-Center/Karnak	MadinahQA	88.16 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
6	Applied-Innovation-Center/Karnak	AraTrust	88.34 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
7	Applied-Innovation-Center/Karnak	ALRAGE	79.03 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
8	Applied-Innovation-Center/Karnak	ArbMMLU-HT	78.04 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
9	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.1	Average	75.86 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
10	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.1	ALiGhafa	78.72 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
11	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.1	ArabicMMLU	75.32 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
12	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.1	EXAMS	58.85 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
13	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.1	MadinahQA	76.82 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
14	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.1	AraTrust	89.68 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
15	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.1	ALRAGE	77.65 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
16	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.1	ArbMMLU-HT	73.96 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
17	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.2	Average	75.85 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
18	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.2	ALiGhafa	78.02 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
19	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.2	ArabicMMLU	75.22 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
20	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.2	EXAMS	58.85 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
21	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.2	MadinahQA	77.01 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/
22	deep-analysis-research/D2IL-Arabic-Qwen2.5-72B-Instruct-v0.2	AraTrust	89.45 %		Open Arabic LLM Leaderboard v2	https://oaii-open-arabic-llm-leaderboard.hf.space/

Model	Benchmark	Score	Unit	Source	Source URL
Qwen/Qwen3.5-397B-A17B-FP8	AVERAGE	68.06 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	ArADICE-Culture	82.78 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	ArabicMMLU	77.54 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	ArabicCulture	61.75 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	PALMX	83.91 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	3LM STEM	88.67 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	AraTrust	90.04 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	MizanQA	73.36 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	MedArabiQA	47.3 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	ArabiLegalQA	54.94 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	GAT	55.89 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	MedArabiBench	47.97 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	HumanEval+	67.68 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	MBPP+	76.72 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Qwen/Qwen3.5-397B-A17B-FP8	FannOrFlop	44.33 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	AVERAGE	66.2 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	ArADICE-Culture	73.33 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	ArabicMMLU	80.94 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	ArabiCulture	53.49 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	PALMX	81.4 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	3LM STEM	93.1 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	AraTrust	89.08 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	MizanQA	55.92 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	MedArabiQA	55.78 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	ArabiLegalQA	71.58 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	GAT	61.06 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	MedArabiBench	54.19 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	HumanEval+	33.54 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	MBPP+	64.35 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Applied-Innovation-Center/Karnak	FannOrFlop	58.91 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Inceptional/jais-2-70B-Chat	AVERAGE	65.81 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
Inceptional/jais-2-70B-Chat	ArADICE-Culture	78.89 %		QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard

This screenshot illustrates the unstructured nature of the original dataset before Power Query (M Language) transformations. It highlights the presence of missing values (nulls), unformatted text, and a denormalized flat-file structure that required extensive ETL processing.

8.3 Core Development Method by Track

Track	Possible Technical Approach
Data Analytics & Data Engineering	ETL/ELT Pipelines & Data Quality: Using Power Query (M Language) for data extraction, deduplication, and applying strict Zero-Imputation logic. Data Warehousing & Integration: Designing a Star Schema architecture (Fact and Dimension tables) optimized for the VertiPaq engine. Business Intelligence & Analytics: Developing automated reporting and dynamic KPIs using 38 custom DAX measures for interactive dashboards.

8.4 DAX Measures Dictionary

The analytical engine of this project is powered by 38 custom DAX measures, strategically grouped into logical folders to handle specific analytical requirements:

01 Core: Includes measures such as Total Models, Total Citations, and Cross-sector Share to define the overall size and collaboration within the dataset.

02 Scale: Includes Median Parameters, Max Parameters, and Known Size, which are used to evaluate the technical size of models and manage data coverage gaps.

03 Accessibility: Includes % Open Weights and Open Training Code Share to accurately measure the true open-source availability of AI models.

04 Time Intelligence: Includes Models YoY %, Models CAGR (5Y), YTD, and 3Y Moving Avg to track exponential growth, rolling averages, and dynamic period-over-period comparisons.

05 Concentration: Includes HHI by Country, HHI by Organization, and CR4 Share to calculate market monopoly indices and league tables, ultimately proving compute dependency.

06 & 07 Regional: Includes Arab Models, Arabic Focused, and Egypt Global Rank, which serve as specialized measures to track regional readiness and evaluate Egypt's specific AI ecosystem.

08 Data Quality: Includes Confident Records, % Unknown Access, and Coverage % acting as operational metrics to ensure dashboard users are aware of data gaps and unrecorded values.

9. Tools and Technologies

Category	Examples
Programming Languages	DAX (Data Analysis Expressions), M-Language (Power Query), HTML/CSS (For Website Integration)
Data Engineering & ETL Tools	Microsoft Power Query
BI & Analytics Platform	Microsoft Power BI Desktop, Power BI Service
Databases / Data Storage	VertiPaq In-Memory Engine, Flat Files (CSV / Excel)
Frontend / Visualization	Power BI Visual Canvas, Web Interface
Version Control	GitHub

11. Data / System Requirements

Requirement Type	Description
Data Requirements	Over 3,500 historical and modern AI model records (1957-2026). Sourced from public databases (e.g., Epoch AI) in CSV/Excel formats. The data requires strict ETL processing, deduplication, and Zero-Imputation to ensure high quality for DAX calculations.
Functional Requirements	Automated calculation of market concentration (HHI), time intelligence tracking (CAGR, YoY), dynamic cross-filtering, and automated Zero-Imputation handling across 6 interactive dashboard pages.
Non-Functional Requirements	High performance with sub-second visual refresh rates (via VertiPaq engine), scalable Star Schema architecture to accommodate future data, and an intuitive UI/UX tailored for executive policymakers.
Hardware Requirements	Standard computing workstation for Power BI Desktop development (minimum 8GB RAM, multi-core CPU recommended). No specialized external hardware, sensors, or heavy servers are required.
Software Requirements	Microsoft Power BI Desktop, Power BI Service, Power Query (M-Language), DAX engine, HTML/CSS for web integration, and standard web browsers for end-user consumption.
Security Requirements	Secure access control and workspace authentication managed via Microsoft Power BI Service, ensuring that strategic reports are shared only with authorized stakeholders.

12. Expected Deliverables

Based on the Data Analytics and Business Intelligence scope of this project, the following final deliverables will be submitted to the evaluation committee:

- **Final Project Documentation:** This comprehensive document detailing the problem statement, system architecture, data dictionary, and technical approach.
- **Data Pipeline & Semantic Model:** A fully functional Power Query (M) ETL pipeline and a highly optimized Star Schema deployed within the VertiPaq engine.
- **Dashboard & Visualization Interface:** A 6-page interactive executive Power BI dashboard powered by 38 custom DAX measures (covering HHI, CAGR, and Arab AI Readiness).
- **Web Interface (Application):** A dedicated project website designed to host the analytical findings, document the methodology, and present the final insights professionally.
- **Testing and Evaluation Output:** Documented data profiling results, Zero-Imputation validation, and DAX mathematical accuracy checks.
- **Final Presentation & Demo Video:** A structured PowerPoint presentation and a recorded demonstration video explaining the system workflow and strategic insights for the Digilians initiative committee.

13. Evaluation and Testing

Explain how the project will be tested and evaluated. These are examples only; each team should select methods that match project objectives.

Project Type	Possible Evaluation Methods
Data Engineering (ETL & Modeling)	Data quality profiling, Zero-Imputation validation, Power Query pipeline success rate, and Star Schema relationship integrity (verifying 1-to-many filters).
Data Analytics & BI (Dashboards)	DAX mathematical accuracy testing (verifying HHI and CAGR outputs manually against raw data), cross-filtering behavior validation, and VertiPaq engine performance evaluation (ensuring sub-second visual refresh rates).
Web Development (Interface)	Usability testing, functional UI testing, and responsive design validation to ensure seamless integration of the analytical documentation.

14. Innovation and Added Value

The added value of this project is represented in delivering a highly centralized, interactive, and automated analytical platform that evaluates global AI compute dependency. Compared with existing approaches—which heavily rely on fragmented public datasets, manual spreadsheet data cleansing, and static reporting—the proposed solution provides significant improvements in data accuracy, processing speed, and decision-making velocity by using a robust Star Schema data model, an engine of 38 custom DAX measures, a Zero-Imputation ETL strategy, and a dedicated web platform. This can help improve national AI readiness evaluations, reduce the risks of critical data fragmentation and manual calculation errors, and directly support strategic decision-makers and the Digilians initiative in achieving the digital transformation goals of Egypt Vision 2030.

To further clarify the project's unique contributions, the innovation and added value can be detailed across four main pillars:

- **1. Technical Automation (Zero-Imputation Strategy):** Shifting from slow, manual data handling to a fully automated Power Query (M-Language) pipeline. The project innovates by applying a strict "Zero-Imputation" logic to handle missing data (Nulls) in over 3,500 records, ensuring that DAX mathematical iterators function flawlessly without dropping critical historical data.
- **2. Advanced Analytical Engine (Market Concentration HHI):** Moving beyond basic descriptive charts (like simple bar or pie charts) by engineering complex,

dynamic DAX measures. A key innovation is automating the Herfindahl-Hirschman Index (HHI) and Time Intelligence metrics to dynamically monitor market monopoly versus open-source democratization in the global AI ecosystem.

- **3. Strategic & Regional Focus:** Providing a dedicated analytical focus on "Arab & Egypt AI Readiness" compared to global computational trends. This transforms abstract global data into actionable, localized intelligence tailored for national tech investors and policymakers.
- **4. Enhanced Accessibility & Web Integration:** Elevating the traditional BI reporting experience by integrating the enterprise-grade Power BI dashboard (spanning 6 analytical pages) with a dedicated web interface. This ensures that complex computational data is highly accessible, interactive, and user-friendly for non-technical stakeholders.

15. Team Roles and Responsibilities

Define the role of each team member. Roles should be adjusted based on the project track and team size.

Team Member	Role	Responsibilities
Nermeen Ahmed	Team Leader & Data Analyst	Leadership & Analysis: Acted as the Team Leader and handled the data calculations. Shared Tasks: Collaborative data cleaning, creating a simple local web page for presentation, and final project review.
Mariam Samir	Data Analyst, Documentation & Presentation	Data, Docs & Pitch: Collected the raw data, co-authored the project documentation, and designed the final presentation. Shared Tasks: Collaborative data cleaning, creating a simple local web page for presentation, and final project review.
Shahd Mohamed	BI Analyst & Dashboard Designer	Design & UI: Structured the data tables and built the Power BI Dashboard visuals. Shared Tasks: Collaborative data cleaning, creating a simple local web page for presentation, and final project review
Amany Mohamed	BI Analyst & Dashboard Designer	Design & UI: Structured the data tables and built the Power BI Dashboard visuals. Shared Tasks: Collaborative data cleaning, creating a simple local web page for presentation, and final project review

Menna Mohamed	Data Analyst & Documentation Member	Analysis & Docs: Co-developed the data calculations and contributed to the project documentation. Shared Tasks: Collaborative data cleaning, creating a simple local web page for presentation, and final project review.
Shrouq Mera	Data Analyst & Poster Lead Member	Data Analyst Analysis & Docs: Executed Power Query ETL data cleaning, compiled the project documentation, and designed the project poster. Shared Tasks: Collaborative data cleaning, creating a simple local web page for presentation, and final project review

16. Ethical, Security, and Privacy Considerations

This project is developed with a strong commitment to data integrity, transparency, and secure access. Given that the solution is a strategic Business Intelligence platform rather than an application handling personal user data, the focus is placed on the following key areas:

- **Using Data Legally and Ethically:** The project strictly relies on public, verifiable data sources (e.g., the Notable AI Systems database). The data is utilized objectively to analyze global computing trends without altering historical records, ensuring full compliance with open-data usage terms and ethical reporting standards.
- **Ensuring Transparency in Results and Decisions:** To prevent misleading strategic decisions, the project emphasizes complete transparency. The methodology behind data cleansing (specifically the Zero-Imputation logic) and the exact mathematical formulas used in the DAX engine (such as the HHI calculation) are fully documented. This ensures stakeholders can trust how market concentration and AI readiness metrics are derived.
- **Applying Secure Authentication and Access Control:** Strategic national insights require secure distribution. The analytical platform leverages Microsoft Power BI Service's robust security infrastructure, implementing workspace access controls and authentication protocols to ensure that the executive dashboards are restricted to authorized policymakers and committee members.
- **Securing Databases and Web Interfaces:** The underlying data architecture (Star Schema) and the integrated project website follow standard web and data security practices to prevent unauthorized modifications to the centralized semantic model and protect the integrity of the published documentation.

17. Expected Impact

Explain the expected benefit of the project. The project is not required to cover all impact areas. These are examples to guide the team; select only what is relevant.

Impact Area	Example Questions
Business / Organizational Impact	Provides executive stakeholders and policymakers with an interactive, data-driven platform to monitor global AI trends, shifting from manual reporting to instant, evidence-based decision-making.
Technical Impact	Establishes a highly automated data pipeline (ETL) and a robust data model, eliminating manual data cleansing efforts and ensuring accurate, real-time metric calculations.
Economic & Strategic Impact	Supports national initiatives, including the Digilians initiative and Egypt Vision 2030, by highlighting "Arab & Egypt AI Readiness," which helps direct technology investments effectively.
User Impact	Offers a user-friendly local web page and an intuitive BI dashboard, making complex AI market data easily accessible and understandable for non-technical users.
Social / Cultural Impact	Raises societal awareness about "Digital Sovereignty" by highlighting the global monopoly of AI technologies. The project underscores the critical need for developing localized AI models to protect regional cultural identities from the ideological biases embedded in foreign AI ecosystems.

18. Conclusion

- **Summary of Core Achievements** In conclusion, this capstone project successfully designed and delivered a comprehensive, end-to-end Business Intelligence (BI) platform dedicated to analyzing global AI computational trends and market concentration. The team successfully transitioned over 3,500 rows of complex, scattered raw data into a highly centralized, interactive, and automated analytical system. Through rigorous teamwork, the project effectively mapped the historical evolution of AI models and provided a clear lens on the current state of "Arab and Egypt AI Readiness."
- **Fulfilling the Core Objectives** The primary problem stated at the beginning of this project—the reliance on fragmented public datasets, manual data cleansing, and static reporting—has been completely resolved. The delivered solution automates the calculation of critical metrics, such as the Herfindahl-Hirschman Index (HHI), saving significant time and eliminating human error. Furthermore, the project successfully addressed the broader strategic objective by highlighting the crucial concept of "Digital Sovereignty," empowering local decision-makers to evaluate cultural and technological biases in foreign AI ecosystems.
- **Reflection on Technical Integration** The strength of this final product lies in the seamless integration of its technical modules. The journey began with data collection and an automated ETL pipeline (Data Engineering) that effectively cleaned and structured the data. This clean data was then fed into a highly optimized Star Schema model (Data Modeling), which served as the foundation for our custom calculations and KPIs (Data Analysis). Ultimately, these hidden technical layers were brought to life through a visually engaging 6-page Power BI dashboard and integrated into a dedicated local web page (User Interface). Every component communicates smoothly to ensure high performance and an intuitive user experience.
- **Final Readiness and Value Delivered Today**, the system is fully functional, rigorously reviewed by all team members, and ready for immediate deployment. It delivers substantial strategic value by providing executive stakeholders within the Digilians initiative with the clear, evidence-based insights required to direct future technology investments. Ultimately, this project stands as a practical tool to support the digital transformation and technological independence goals of Egypt Vision 2030.

19. Future Work

While the current capstone project successfully delivers a comprehensive and fully functional BI platform, there is a strategic roadmap for future enhancements. These improvements can be categorized into three main areas:

1. Data Expansion & Analytical Depth:

Expand Dataset Size and Regional Coverage: Future iterations aim to include granular datasets focusing specifically on MENA region AI startups and local computational infrastructures, providing a deeper dive into Arab AI readiness.

Add Predictive Analytics: Expanding the analytical scope by introducing forecasting features to estimate future AI compute requirements and market concentration trends (HHI) over the next 5 to 10 years.

2. Accessibility & User Experience (UI/UX):

Add Multilingual Support: Implementing full Arabic localization for both the Power BI dashboard and the web interface to better serve local policymakers and the broader Arab community.

Add Mobile Application Support: Developing a mobile-optimized view of the dashboard, allowing executive stakeholders to access real-time AI market insights seamlessly from their smartphones.

Automated Stakeholder Alerts: Setting up automated email notifications to alert decision-makers whenever global AI monopoly indicators exceed certain thresholds.

3. System Integration & Security:

Integrate with External Systems: Linking the project's web interface directly with official national platforms, such as the Digilians portal or Egypt Vision 2030 digital archives.

Real-Time API Integration: Transitioning from scheduled data imports to real-time API connections with global AI databases, ensuring the dashboard automatically reflects the absolute latest market shifts without manual data refreshes.

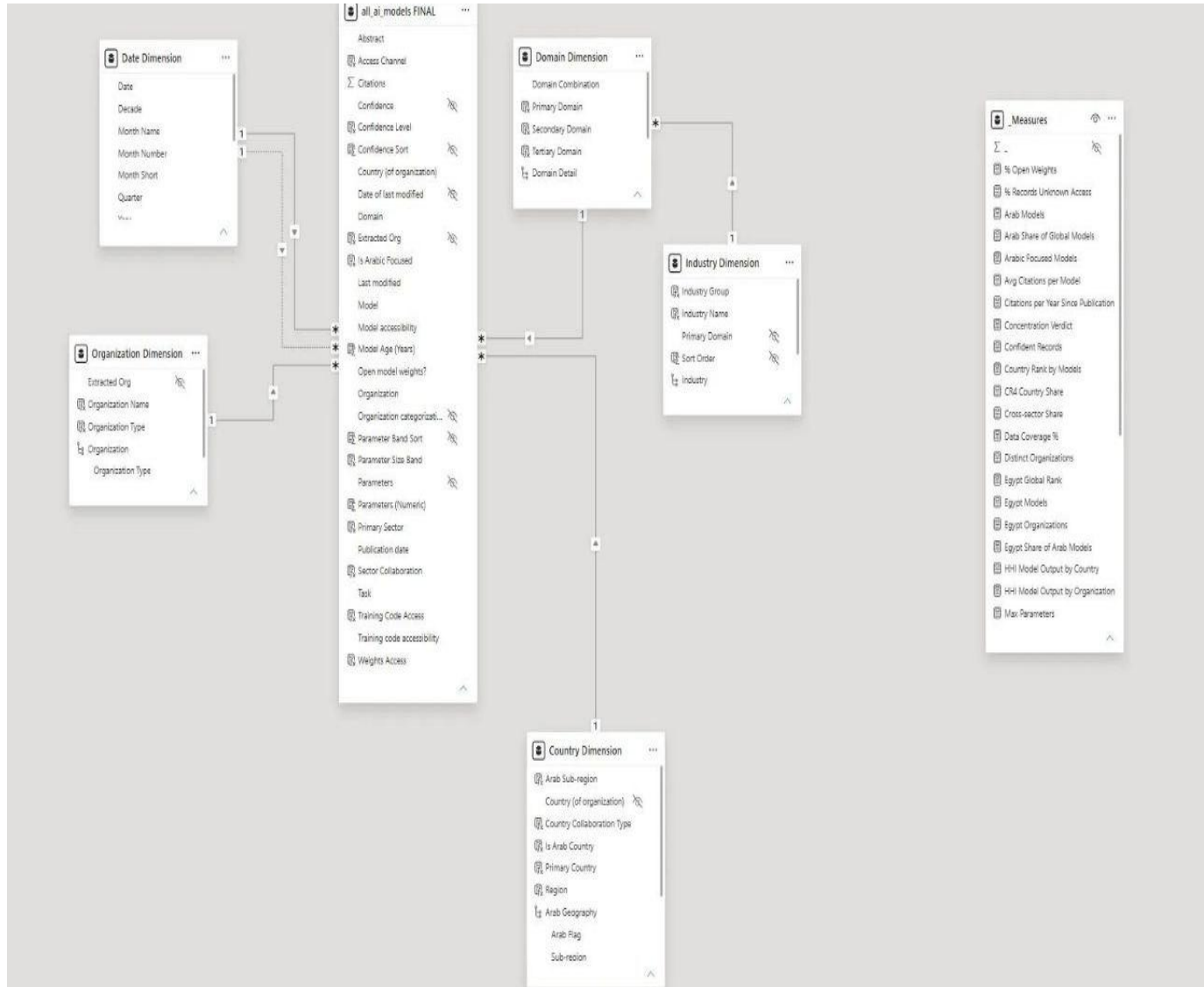
Improve Security Features (Row-Level Security): Implementing advanced access controls (RLS) within Power BI so that different user groups (e.g., investors vs. researchers) only see the data and reports relevant to their specific clearance levels.

20. References / External Resources (Optional)

- the following external resources, datasets, and official documentation were utilized to develop the system, ensure data accuracy, and formulate the analytical methodologies:
- Epoch AI Database (Primary Dataset): Comprehensive database of notable artificial intelligence and machine learning systems (1957–2026), used for analyzing global compute trends. Link: <https://epochai.org/data/notable-ai-models>
- Microsoft Power BI Official Documentation: Used for configuring the VertiPaq engine, implementing the Star Schema data model, and designing the visualization interface. Link: <https://learn.microsoft.com/en-us/power-bi/>
- Data Analysis Expressions (DAX) Reference: Official Microsoft documentation utilized for writing the 38 custom measures and performing dynamic time-intelligence calculations. Link: <https://learn.microsoft.com/en-us/dax/>
- Power Query M Formula Language: Utilized for building the automated ETL pipeline and implementing the Zero-Imputation logic for data cleansing. Link: <https://learn.microsoft.com/en-us/powerquery-m/>
- Herfindahl-Hirschman Index (HHI) Methodology: The mathematical standard and economic methodology used to calculate and assess market concentration and monopoly in the AI compute sector. Link: <https://www.justice.gov/atr/herfindahl-hirschman-index>
- Egypt Vision 2030 (Strategic Alignment): Sustainable Development Strategy utilized as the strategic framework for evaluating "Arab & Egypt AI Readiness." Link: <https://mped.gov.eg/Vision?lang=en>

Appendices

Appendix A: Screenshots / Prototype Design



Model

1	Model	Benchmark	Score	Unit	Source	Source URL
2	Qwen/Qwen3.5-397B-A17B-FP8	AVERAGE	68.06	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
3	Qwen/Qwen3.5-397B-A17B-FP8	ArabiQCE-Culture	82.78	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
4	Qwen/Qwen3.5-397B-A17B-FP8	ArabicMMUJ	77.54	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
5	Qwen/Qwen3.5-397B-A17B-FP8	ArabicCulture	61.75	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
6	Qwen/Qwen3.5-397B-A17B-FP8	PALMX	83.91	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
7	Qwen/Qwen3.5-397B-A17B-FP8	3LM STEM	88.67	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
8	Qwen/Qwen3.5-397B-A17B-FP8	AraTrust	90.04	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
9	Qwen/Qwen3.5-397B-A17B-FP8	MizanQA	73.36	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
10	Qwen/Qwen3.5-397B-A17B-FP8	MedArabiQ	47.3	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
11	Qwen/Qwen3.5-397B-A17B-FP8	ArabiLegalQA	54.94	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
12	Qwen/Qwen3.5-397B-A17B-FP8	GAT	55.89	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
13	Qwen/Qwen3.5-397B-A17B-FP8	MedArabiBench	47.97	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
14	Qwen/Qwen3.5-397B-A17B-FP8	HumanEval+	67.68	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
15	Qwen/Qwen3.5-397B-A17B-FP8	MBPP+	76.72	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
16	Qwen/Qwen3.5-397B-A17B-FP8	FannOrFlop	44.33	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
17	Applied-Innovation-Center/Karnak	AVERAGE	66.2	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
18	Applied-Innovation-Center/Karnak	ArabiQCE-Culture	73.33	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
19	Applied-Innovation-Center/Karnak	ArabicMMUJ	80.94	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
20	Applied-Innovation-Center/Karnak	ArabiCulture	53.49	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
21	Applied-Innovation-Center/Karnak	PALMX	81.4	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
22	Applied-Innovation-Center/Karnak	3LM STEM	93.1	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
23	Applied-Innovation-Center/Karnak	AraTrust	89.08	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
24	Applied-Innovation-Center/Karnak	MizanQA	55.92	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
25	Applied-Innovation-Center/Karnak	MedArabiQ	55.78	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
26	Applied-Innovation-Center/Karnak	ArabiLegalQA	71.58	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
27	Applied-Innovation-Center/Karnak	GAT	61.06	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
28	Applied-Innovation-Center/Karnak	MedArabiBench	54.19	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
29	Applied-Innovation-Center/Karnak	HumanEval+	33.34	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
30	Applied-Innovation-Center/Karnak	MBPP+	64.35	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
31	Applied-Innovation-Center/Karnak	FannOrFlop	58.91	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
32	Inceptional/Jais-2-70B-Chat	AVERAGE	65.81	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard
33	Inceptional/Jais-2-70B-Chat	ArabiQCE-Culture	78.89	%	QIMMA Arabic Leaderboard — April 2026	https://huggingface.co/blog/tiuae/qimma-arabic-leaderboard

Financials

TOP500_Compute

Arab_Investment

Arab_Infra

Energy_Prices

Global_AI_Systems

Chip_Presence

QIMMA_60_Rows

OALL_760_Rows

MASTER_Actual_Dat

localhost:3000/#registry

EGYPT AI SOVEREIGN DECISION ENGINE

v2.4

Priority Sectors

National Model Matrix

Challenge Registry

Infrastructure Fit

Egypt Vision 2030

Register Challenge

Government Login

1. Egypt AI Solver - Home & Strategy

2. Architecture Evaluation Report (Red Palm Weevil)

Low 10/2020 Data Sovereignty Certified

Connecting Field Realities to Sovereign AI Compute

The national engine bridges public sector field problems with optimal local and open-source computing models through rigorous edge-readiness testing, data protection, and classification benchmarks across 7 strategic sectors.

Agriculture & Food Security

Comprehensive Healthcare

Traffic & Urban Mobility

SOVEREIGN DECISION ENGINE

Water Security & Nile Basin

Tourism & Heritage

Manufacturing & Industry

Sovereign AI Principle: Prioritizing local inference, data containment on Egyptian soil, low power consumption, and air-gapped models to guarantee operational continuity, national security, and non-reliance on foreign cloud servers.

NATIONAL PRIORITY REGISTRY

7 Priority Strategic Sectors

Registered institutional challenges matched with verified, deployable sovereign AI models.

14 Challenges

10 Challenges

8 Challenges

Submit New Sector Bottleneck

Arab Republic of EgyptNational Council for Artificial Intelligence (NCAI) — Cabinet of Egypt

EGYPT AI SOVEREIGN DECISION ENGINE

National Platform for Sovereign AI Deployment & Edge Viability

v2.4

Priority Sectors

National Model Matrix

Challenge Registry

Infrastructure Fit

Egypt Vision 2030

Register Challenge

Government Login

1. Egypt AI Solver - Home & Strategy

2. Architecture Evaluation Report (Red Palm Weevil)

Law 151/2020 Data Sovereignty Certified

Home / Strategic Sectors / Agriculture & Food Security / Challenge #AG-01 / Architecture Evaluation Report

Direct Production Ready

Challenge ID: AG-01

Egypt Vision 2030 Aligned

Sovereign Architecture Evaluation Report: Red Palm Weevil & Field Crop Pest Vision Diagnostics

Comparative edge feasibility, operational latency, accuracy, and data sovereignty compliance report for handheld and drone edge vision diagnostics across Egyptian date palm plantations (New Valley, Toshka, Aswan).

RFP Specification Document (PDF)

Register Interest

1. National Proposed Models Matrix (Ranked by Field Priority & Viability)

Evaluation Date: Q3 2026

TIER 1: RECOMMENDED SOVEREIGN STANDARD

Offline Air-Gap Direct Selection

YOLOv8 Nano / MobileNetV4 (Quantized INT8)

Quantized INT8 Edge Vision Model (4.2M params) - Optimized for handheld NPU and Low-Power Field Drones

Viability Score

98.2 / 100

Field Accuracy / F1

95.4%

45k palm trunk tests

Inference Latency

12-18 ms

RK3588 / PL3 NPU

Sovereign Footprint

2.4 MB

ONNX / NCNN INT8

Cost / 10k Inferences

0.05 EGP

Zero foreign API fees

Offline Field Viability (Desert Heat & Air-Gap)

98%

Hardware Component Availability in Egypt

95%

BENCHMARK EVALUATION PARAMETERS

Target Field Hardware: Rockchip RK3588 / Jetson Orin Nano / Android Arm64 Handhelds

Field Training Dataset: 45,200 date palm trunk & canopy images (Ministry of Agriculture & Toshka Ag. Authority)

Environmental Tolerance: 50°C desert heat, dust storms, direct high-lux solar glare

Network Connectivity Requirement: Strictly Zero (100% Offline Air-Gapped)

DATA SOVEREIGNTY & LAW 151/2020

100% PASSED

Data Stays Exclusively on Local Device

Arab Republic of EgyptNational Council for Artificial Intelligence (NCAI) — Cabinet of Egypt

EGYPT AI SOVEREIGN DECISION ENGINE

National Platform for Sovereign AI Deployment & Edge Viability

v2.4

Priority Sectors

National Model Matrix

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Law 151/2020 Data Sovereignty Certified

Official National AI Decision Engine — Arab Republic of Egypt

From National Challenges to Sovereign AI: Localizing Decisions, Protecting Data

A national decision-support engine matching public sector field challenges across food security, water, healthcare, logistics, and heritage to verified, edge-deployable, high-security, local and open-source models with zero external data sovereignty leaks and resource efficiency.

Register Challenge & Benchmark Model

Explore 7 Priority Sectors →

View Red Palm Weevil Evaluation Report →

7

Priority Sectors

14

Priority Challenges

42

Benchmark Models

98%

Sovereignty Score

NATIONAL SOVEREIGN EDGE MATRIX

AIR- GAP VERIFIED

Offline / Edge Deployment Ready

92%

Zero Cloud Vendor Lock-In

100%

Data Protection Law 151/2020 Compliance

PASSED

FEATURED PRODUCTION BENCHMARK

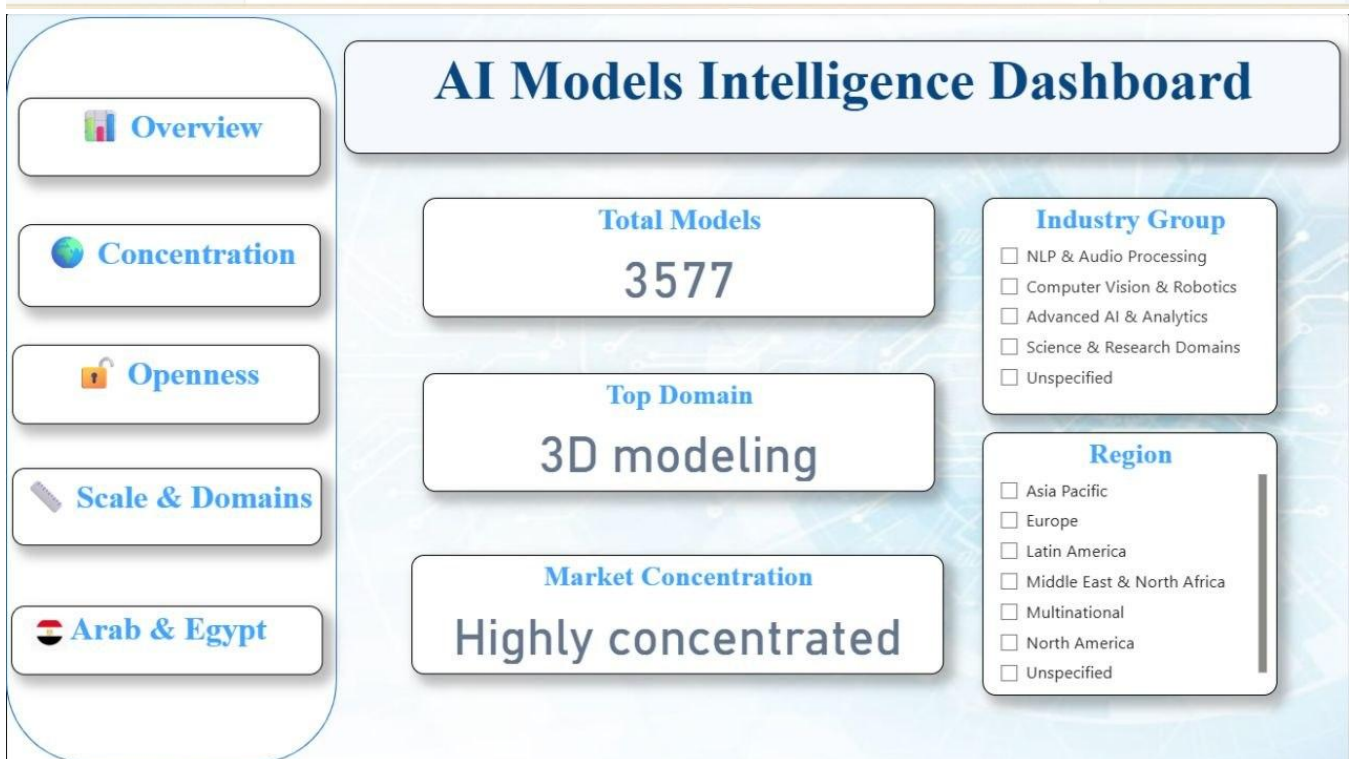
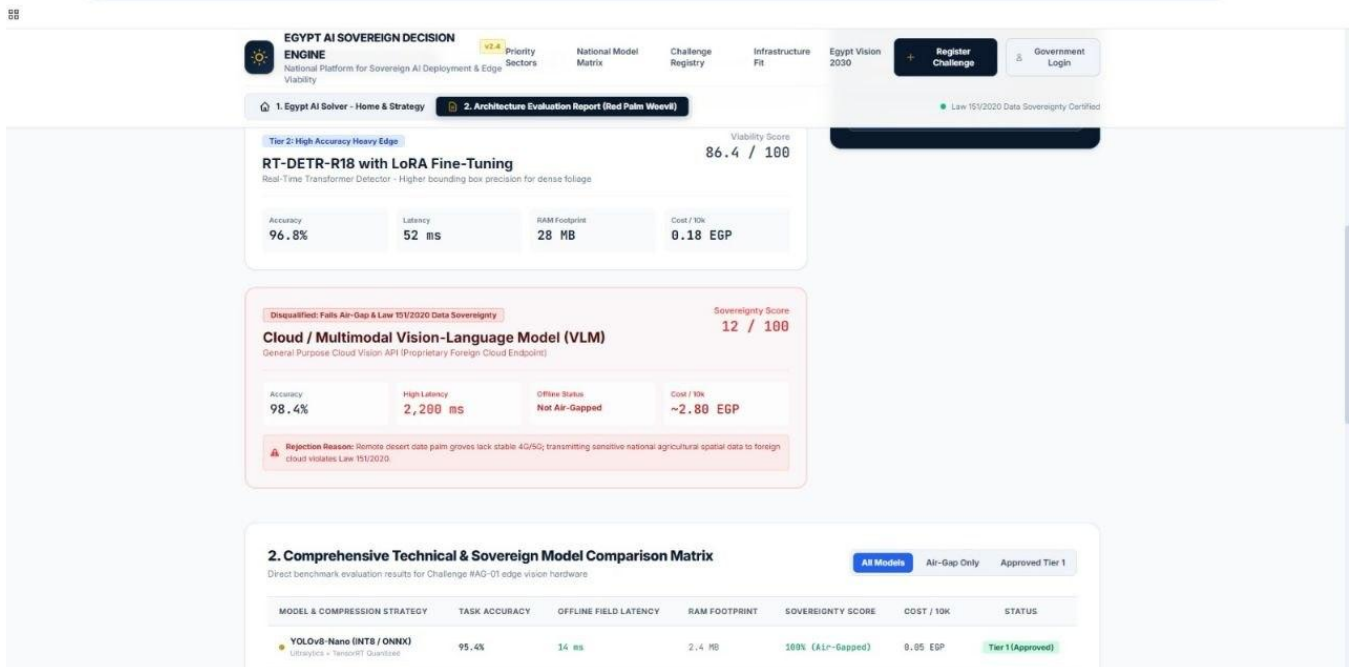
#AG-01: Red Palm Weevil Edge Vision

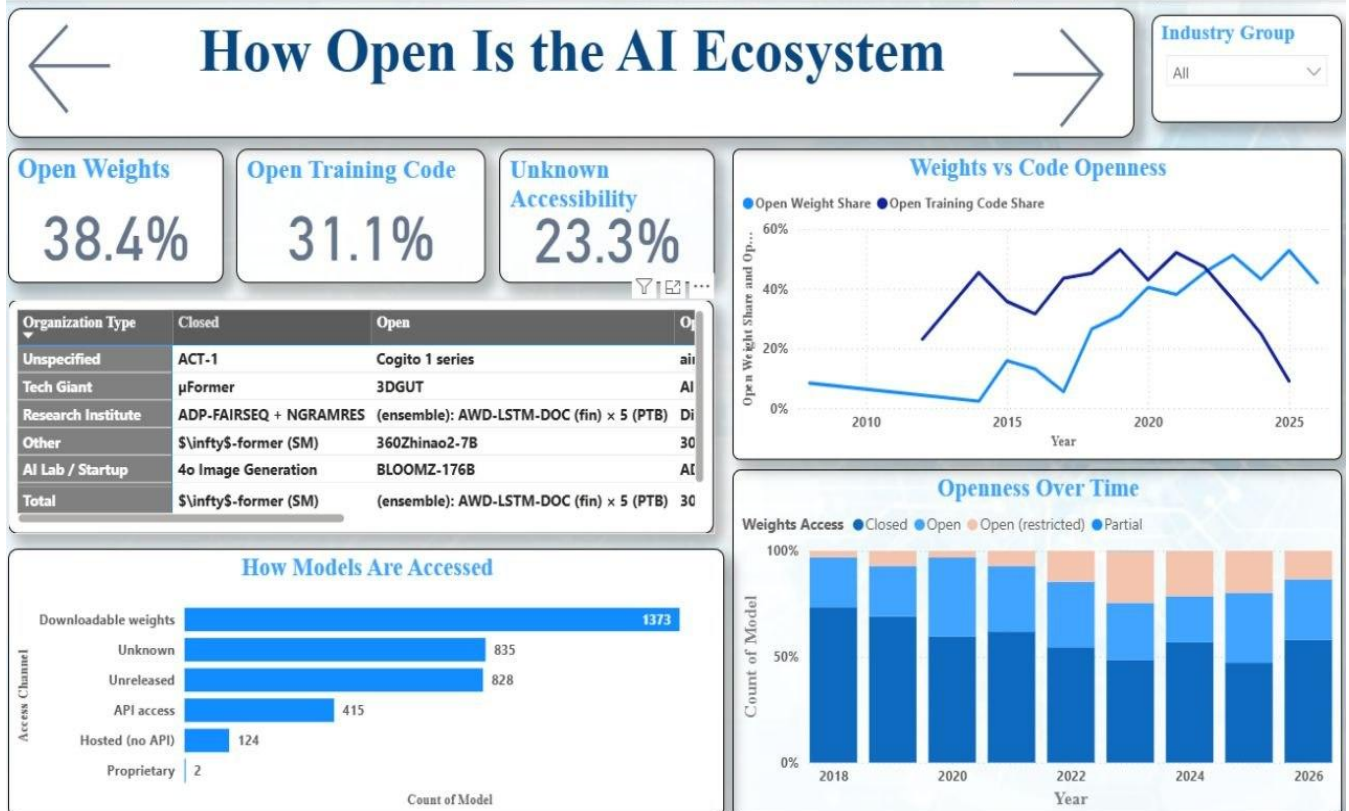
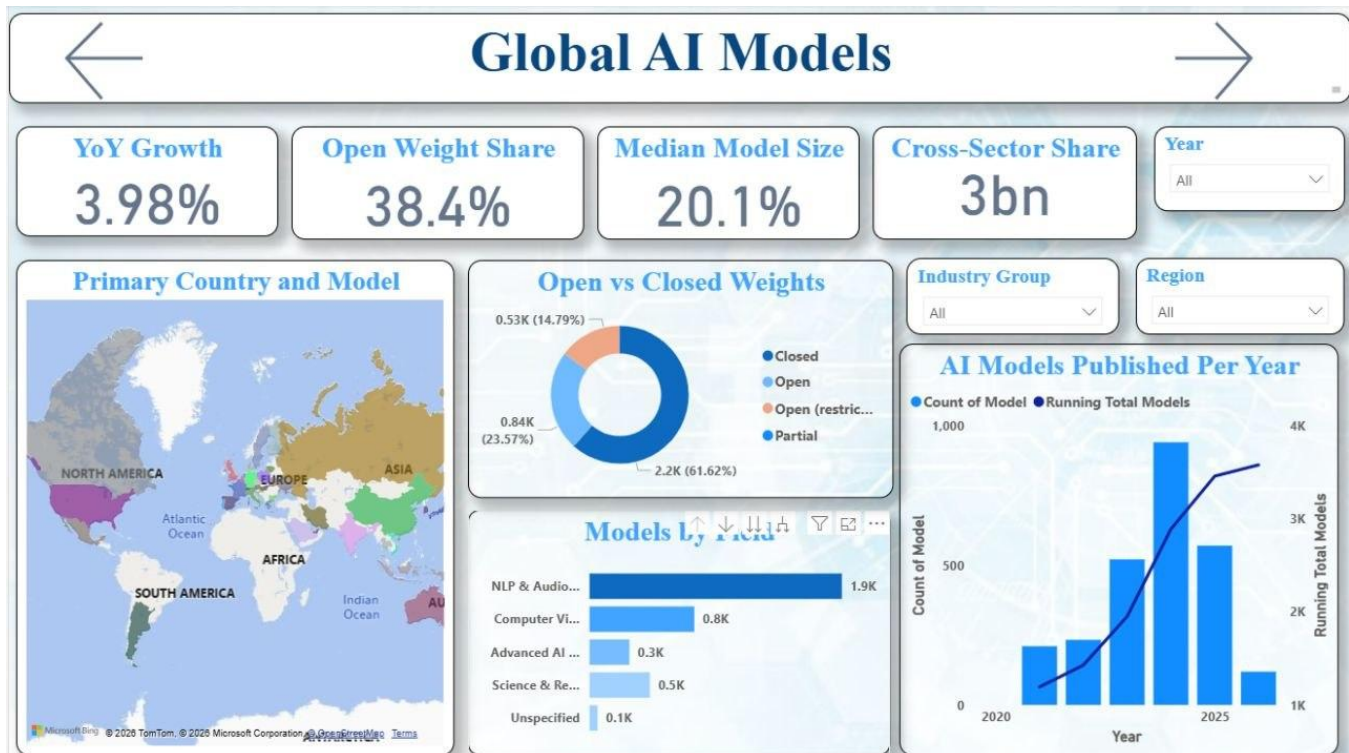
12 ms / RK3588

Deployed across Toshka & New Valley date palm plantations with INT8 Quantized YOLOv8/MobileNetV4.

Open Architecture Evaluation Report →

35







The Arab World & Egypt in AI

Arab Models

32

Arab Share Of Global Models

0.89%

Arabic Focused Models

38

Egypt Global Rank

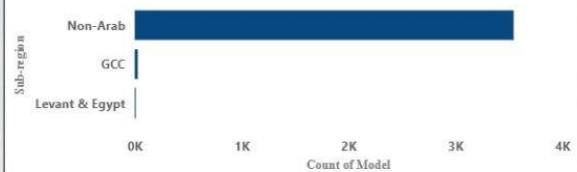
Not ranked

Industry Group

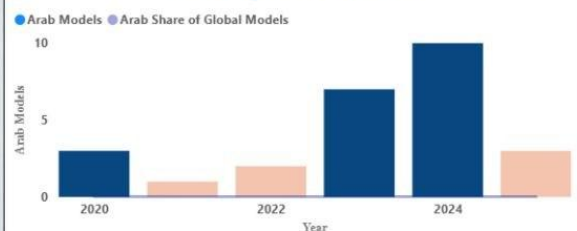
All

Model	Organization Name	Primary Country	Publication date
AlexaTM 20B	Amazon	United States	Tuesday, Aug 14, 2021
ALLaM 34B	Saudi Data and Artificial Intelligence Authority	Saudi Arabia	Tuesday, May 18, 2022
ALLaM 7B	Saudi Data and Artificial Intelligence Authority	Saudi Arabia	Tuesday, May 18, 2022
ALLaM adapted13B	Saudi Data and Artificial Intelligence Authority	Saudi Arabia	Tuesday, May 18, 2022
ALLaM adapted 70B	Saudi Data and Artificial Intelligence Authority	Saudi Arabia	Tuesday, May 18, 2022
ALM 1.0	Beijing Academy of Artificial Intelligence / BAAI	China	Monday, Nov 15, 2022
AraBERT	American University of Beirut	Lebanon	Monday, Mar 27, 2023
AraBERT Large v2	American University of Beirut	Lebanon	Monday, Mar 27, 2023
AraELECTRA	American University of Beirut	Lebanon	Sunday, Mar 26, 2023
AraGPT2-Mega	American University of Beirut	Lebanon	Thursday, Dec 14, 2023
Cohere Embed 4	Cohere	Canada	Tuesday, April 18, 2023
Command R+	Cohere	Canada	Thursday, April 13, 2023
Contriever	Meta AI	United States	Thursday, Dec 14, 2023
Falcon 2 11B	Technology Innovation Institute	United Arab Emirates	Thursday, May 18, 2023
Falcon Mamba	Technology Innovation Institute	United Arab Emirates	Monday, Aug 14, 2023
Falcon-180B	Technology Innovation Institute	United Arab Emirates	Wednesday, Sep 13, 2023
Falcon3-7B	Technology Innovation Institute	United Arab Emirates	Tuesday, Dec 12, 2023
Falcon-40B	Technology Innovation Institute	United Arab Emirates	Wednesday, Nov 15, 2023
Falcon-7B	Technology Innovation Institute	United Arab Emirates	Monday, April 10, 2023
Falcon-Arabic	Technology Innovation Institute	United Arab Emirates	Wednesday, Nov 15, 2023
Falcon-H1	Technology Innovation Institute	United Arab Emirates	Wednesday, Nov 15, 2023
Total			

Count of Model by Sub-region



Arab Output vs Global Share



AI Model Scale & Domain Analysis



Median Model Size

3bn

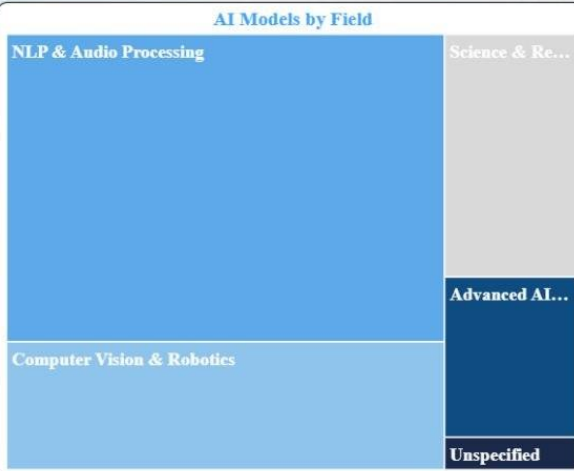
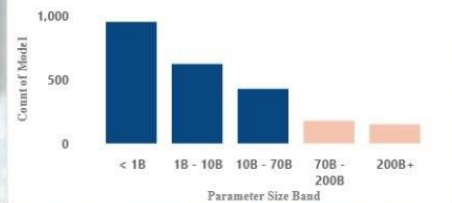
Data Coverage

65.1%

Largest Model

174T

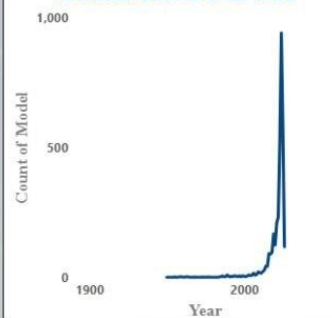
Model Size Distribution



Industry Group

All

AI Fields Growth Over Time



Model Size vs. Research Impact

